

Roll No.:

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B. Tech CSE & SPL

Semester: VI

Name of the Course: Network and System Security

Course Code: TCS 619

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20 Marks)	
(a)	What is the function of routing control as a security mechanism in the OSI model? How does it mitigate risks such as data interception and spoofing?	CO2
(b)	How would you compare chosen plaintext attack and chosen ciphertext attack? Which one is more easy to launch and why?	CO3
(c)	What are the primary goals of the Digital Signature Standard (DSS) algorithm? Demonstrate the signature generation and verification procedures of DSS.	CO5
Q2	(20 Marks)	
(a)	Explain the encryption and decryption procedures of the Vernam cipher. Prove that the Vernam cipher (one-time pad) is unbreakable if the key is truly random, used only once, and kept secret.	CO3
(b)	In the RSA algorithm select two prime numbers, i.e., $p=13$ and $q=11$. Use the RSA algorithm to generate the public and private keys. Further encrypt the plaintext message $M=5$.	CO1
(c)	What is the primary difference between symmetric and asymmetric encryption methods? Explain the trade-offs between performance and security when choosing between symmetric and asymmetric encryption.	CO2
Q3	(20 Marks)	
(a)	Describe the basic components of a network security model. What are the typical threats considered in a network security model?	CO1
(b)	What is S/MIME and what purpose does it serve in email communication? Demonstrate the working procedure of EnvelopedData and SignedData in S/MIME.	CO5
(c)	Explain the architecture of PKIX framework using a figure. How does PKIX ensure trust among entities that have never directly communicated before?	CO6

Q4	(20 Marks)	
(a)	What is a rootkit, and how does it differ from other types of malware? How do rootkits maintain persistence in a compromised system?	CO4
(b)	What is the use of Pretty Good Privacy (PGP)? How PGP provides confidentiality, integrity, and authentication to the exchanged emails?	CO2
(c)	Let two users, Alice and Bob, agree on a public prime number $q=23$ and a primitive root $\alpha=5$. Alice selects a private key $X_A=6$, and Bob selects a private key $X_B=15$ then performs the following calculations. (i) Compute Alice's public key (ii) Compute Bob's public key (iii) Compute the shared secret key that both Alice and Bob derive.	CO1
Q5	(20 Marks)	
(a)	What are the main components of the Secure Sockets Layer (SSL) protocol stack? Describe the operations of SSL record protocol using a figure.	CO1
(b)	Discuss how malware-as-a-service (MaaS) and ransomware-as-a-service (RaaS) are changing the cybercrime landscape.	CO5
(c)	Investigate the working mechanism of the Kerberos authentication process including the role of the Authentication Server (AS), Ticket Granting Server (TGS), and Service Server.	CO6